User Manual - individual work

Interactive Visualization Tool for ISO 15926-4

Overview

This manual covers the features developed to enhance the usability and functionality of the Interactive Visualization Tool for the ISO 15926-4 Standard. The features include UI improvements such as tooltips and buttons, a comparative view for side-by-side data visualization, search history functionality, and layout adjustments to ensure a better user experience. These features are designed to help users explore and interact with hierarchical data in a more intuitive and efficient way.

Installation and Setup

1. Clone the Repository

To set up the tool locally, use the following commands

```
git clone https://github.com/nlp-tlp/15926-4-CITS5206-18.git
cd 15926-4-CITS5206-18
```

2. Install Required Dependencies

```
pip install -r requirements.txt
```

3. Run the Streamlit Application

```
streamlit run src/main.py
```

The application will open in your browser, and you can start exploring the visualizations.

Using the Features

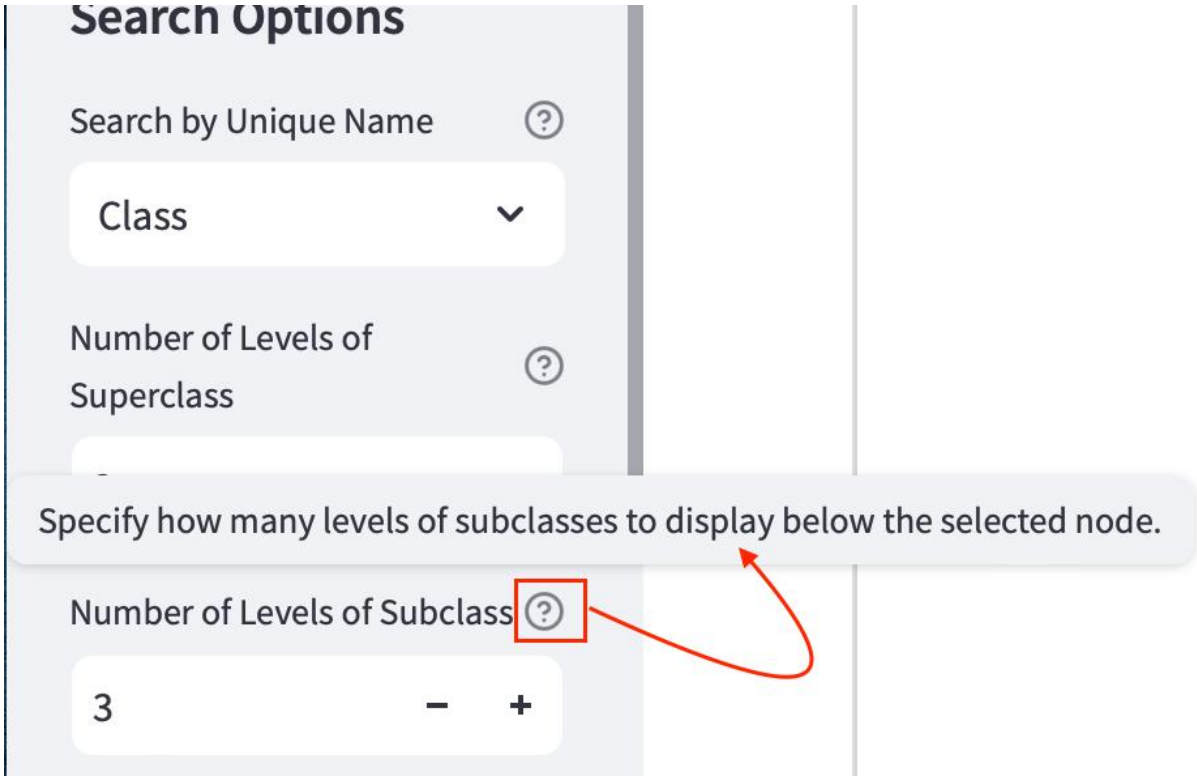
1. Tooltips and Buttons

- ***Feature Description***

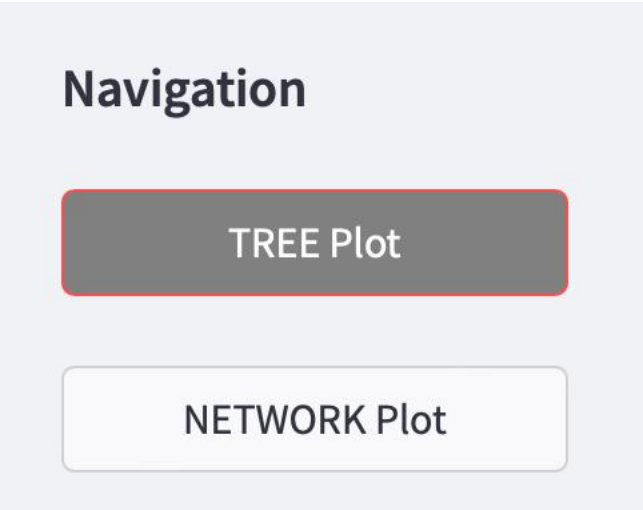
Tooltips provide real-time contextual information about the nodes in the graph, allowing users to see additional details when hovering over a node. The buttons improve user interaction by enabling actions like switching views or selecting nodes directly from the interface.

- ***How to Use***

Tooltips: Hover over a node to see a tooltip that displays the node's description. This feature enhances the user experience by giving immediate feedback without needing to click.



Buttons: When you move your cursor over any button (e.g., Reset, Zoom In/Out, or Adjust Layout), a shadow appears around the button, visually indicating that it is clickable. This feedback helps guide users, especially new users, by showing them where interactive elements are.



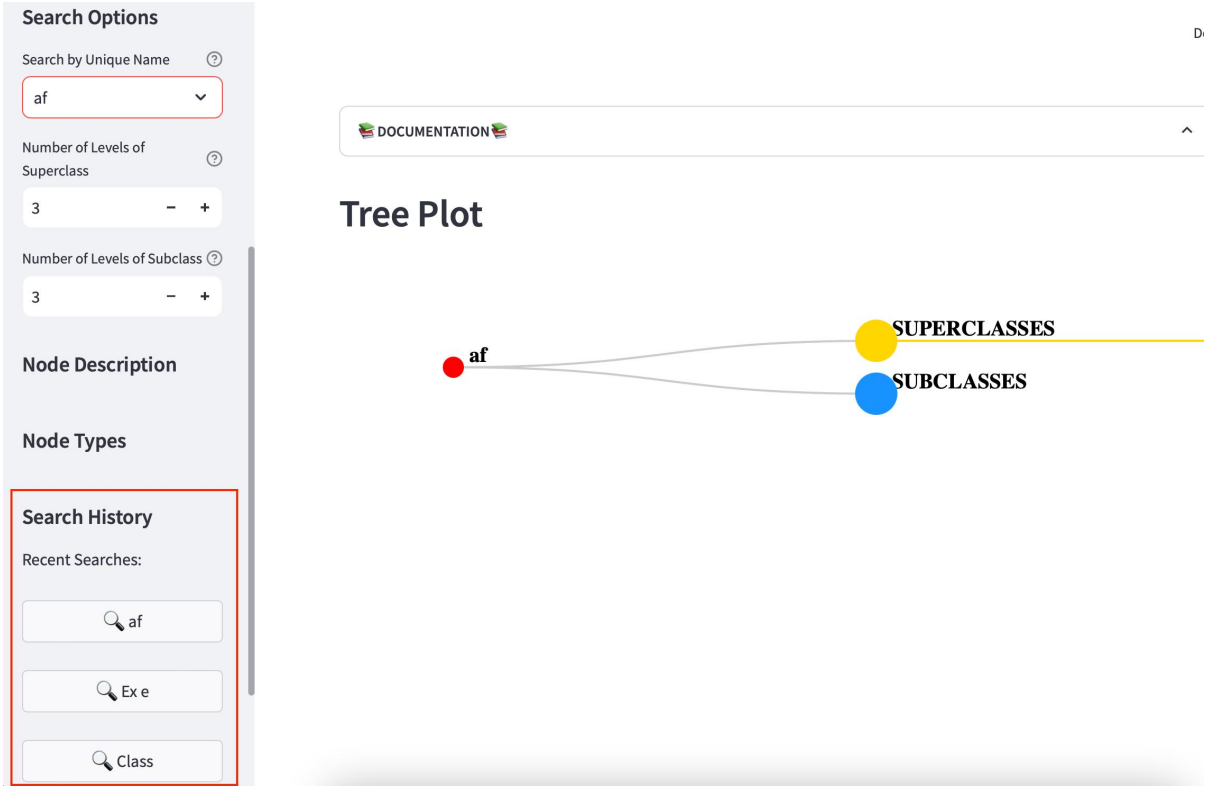
2. Search History Feature

- ***Feature Description***

The search history feature tracks previous searches, allowing users to easily revisit previous queries without re-entering them. This improves workflow efficiency by providing quick access to frequently or recently viewed nodes.

- ***How to Use***

When a search is performed, it is saved in the sidebar under the Search History section. To revisit the previous searches, click on the search name directly in the history. The corresponding graph will be reloaded and displayed. The recent search history will be clear if the web page is refreshed.



3. Comparative View Feature

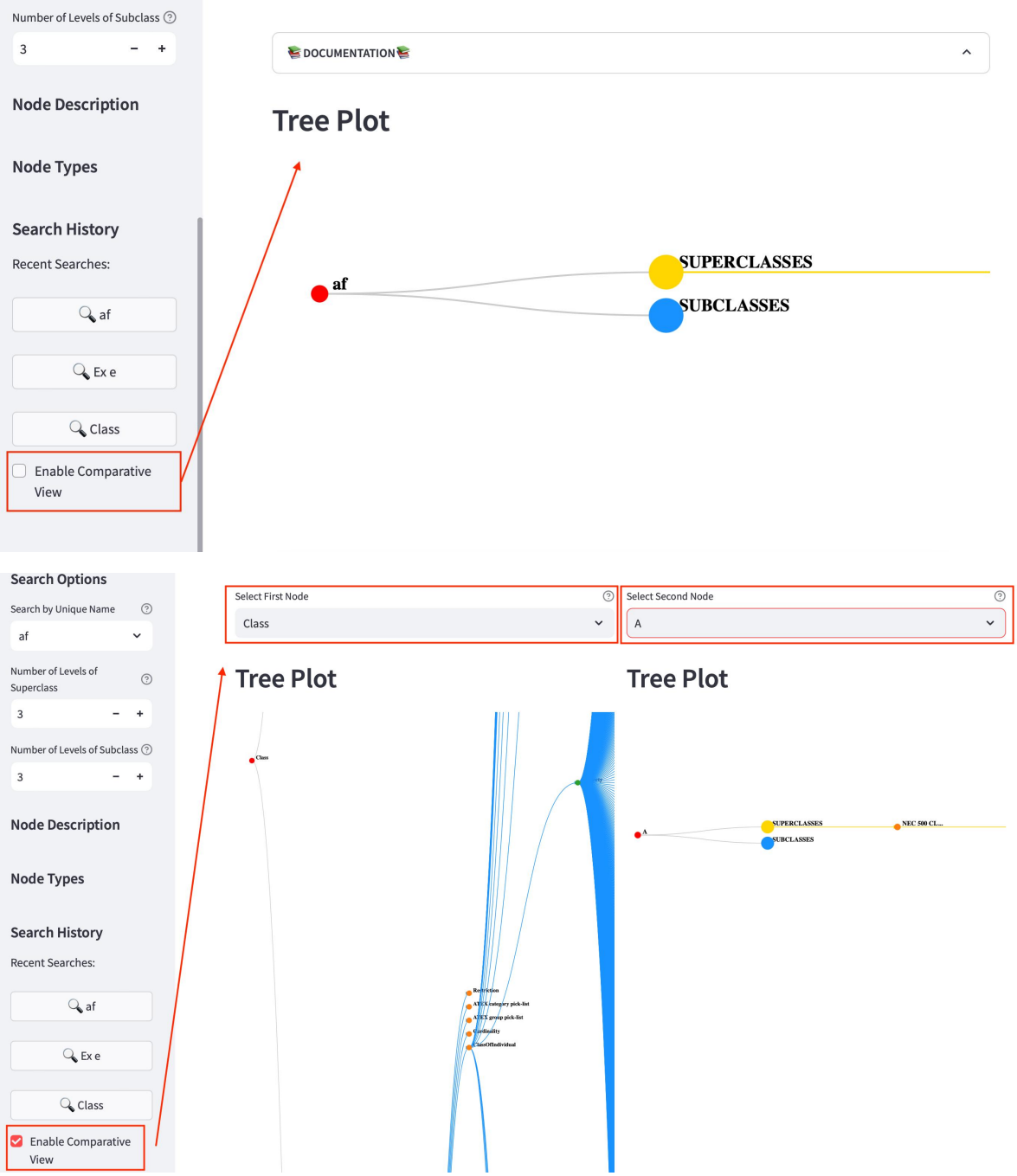
- ***Feature Description***

The comparative view allows users to compare two visualizations side by side. This feature is especially useful for analyzing different hierarchical structures or datasets in the same interface, providing a visual way to compare relationships between nodes.

- ***How to Use***

From the sidebar, select the Enable Comparative View option. And then you can choose two nodes which want to explore and compare. The screen will display two visualizations side by side, allowing you

to explore and compare them simultaneously. Adjust the number of displayed levels for each visualization independently to control the depth of the data being shown.



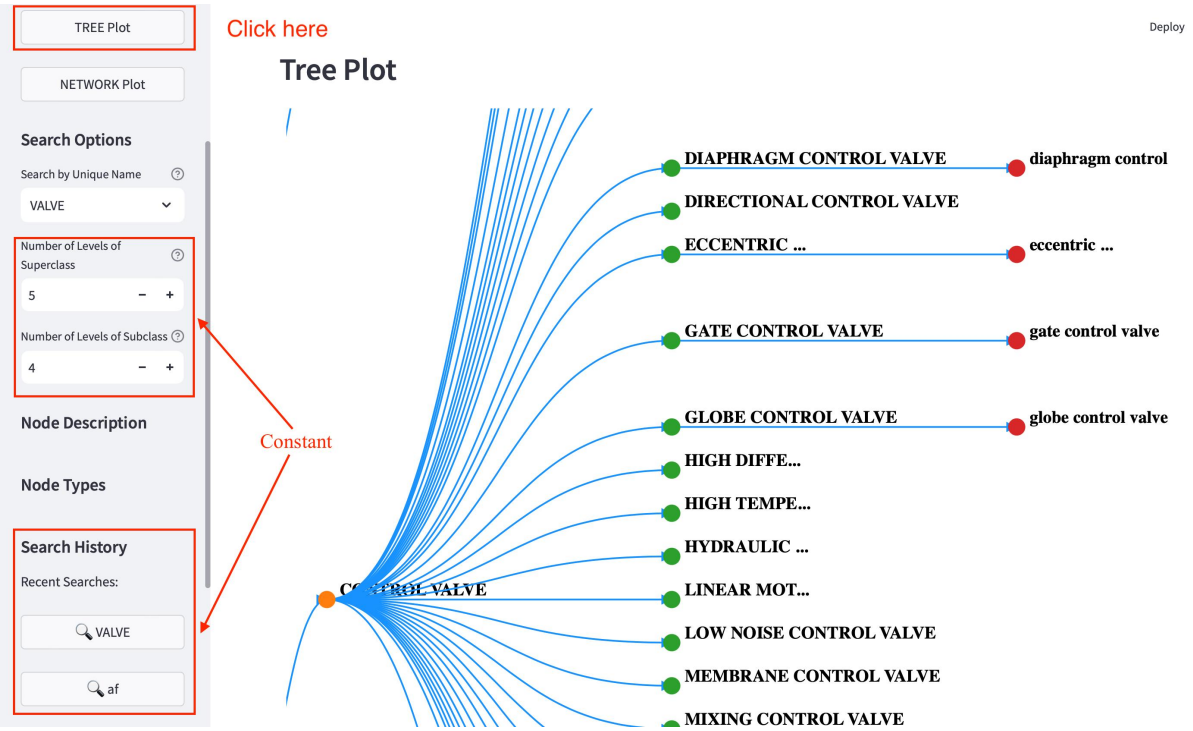
4. Keeping Sidebar Constant When Switching Between Plots

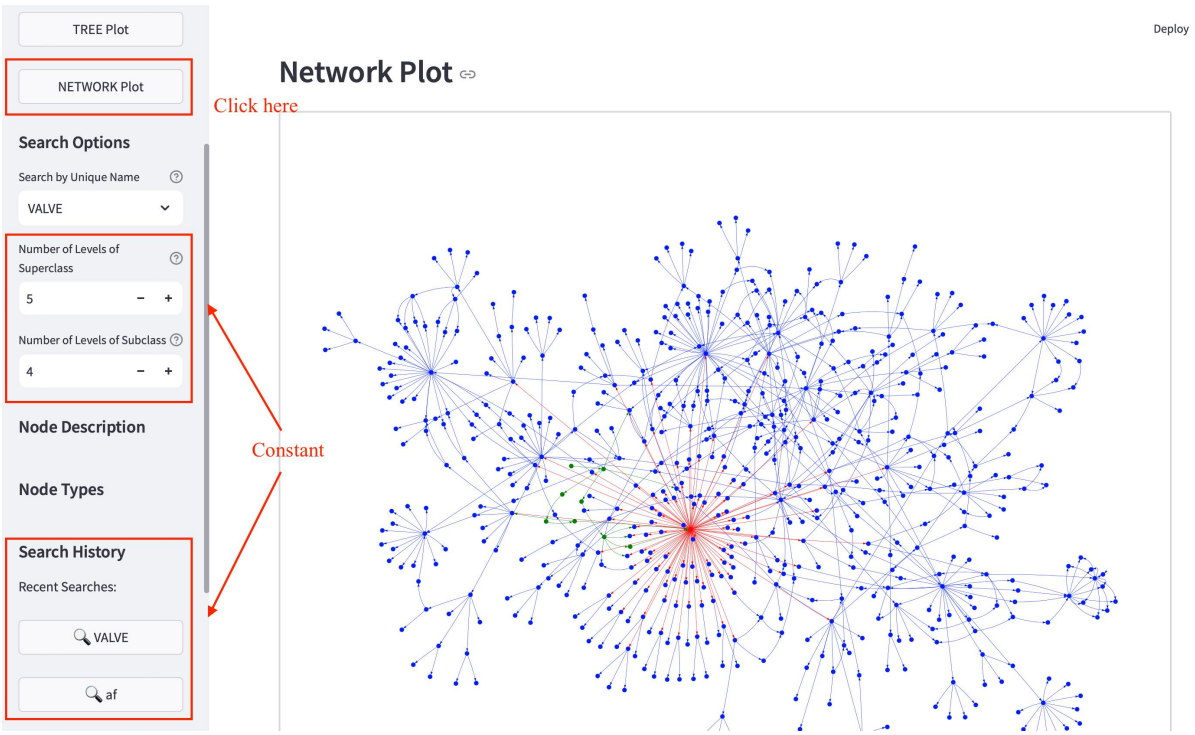
- ***Feature Description***

When users switch between visualizations (e.g., moving from the Network Plot to the Tree Plot), the sidebar retains the following features. Any searches performed remain visible, and the focused node is still highlighted. The number of superclass and subclass levels displayed remains the same even after switching between plots. Previously entered search terms are preserved in the search history, so users can revisit their prior queries.

- ***How to Use***

Use the buttons in the sidebar to switch between different views or to interact with the graph in various ways.





Conclusion

This user manual provides a comprehensive guide to using the features I developed in the Interactive Visualization Tool for the ISO 15926-4 Standard. By following the instructions and tips outlined here, users can fully explore and interact with the hierarchical data through tooltips, comparative views, search history, and improved layout functionality.